Corner Detection

Shree K. Nayar

Columbia University

Topic: Edge Detection, Module: Features

First Principles of Computer Vision

© 2020 Shree K. Nayar

Corners

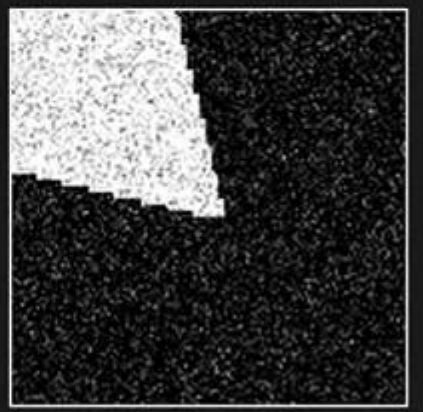
Corner: Point where Two Edges Meet. i.e., Rapid Changes of Image Intensity in Two Directions within a Small Region



"Flat" Region

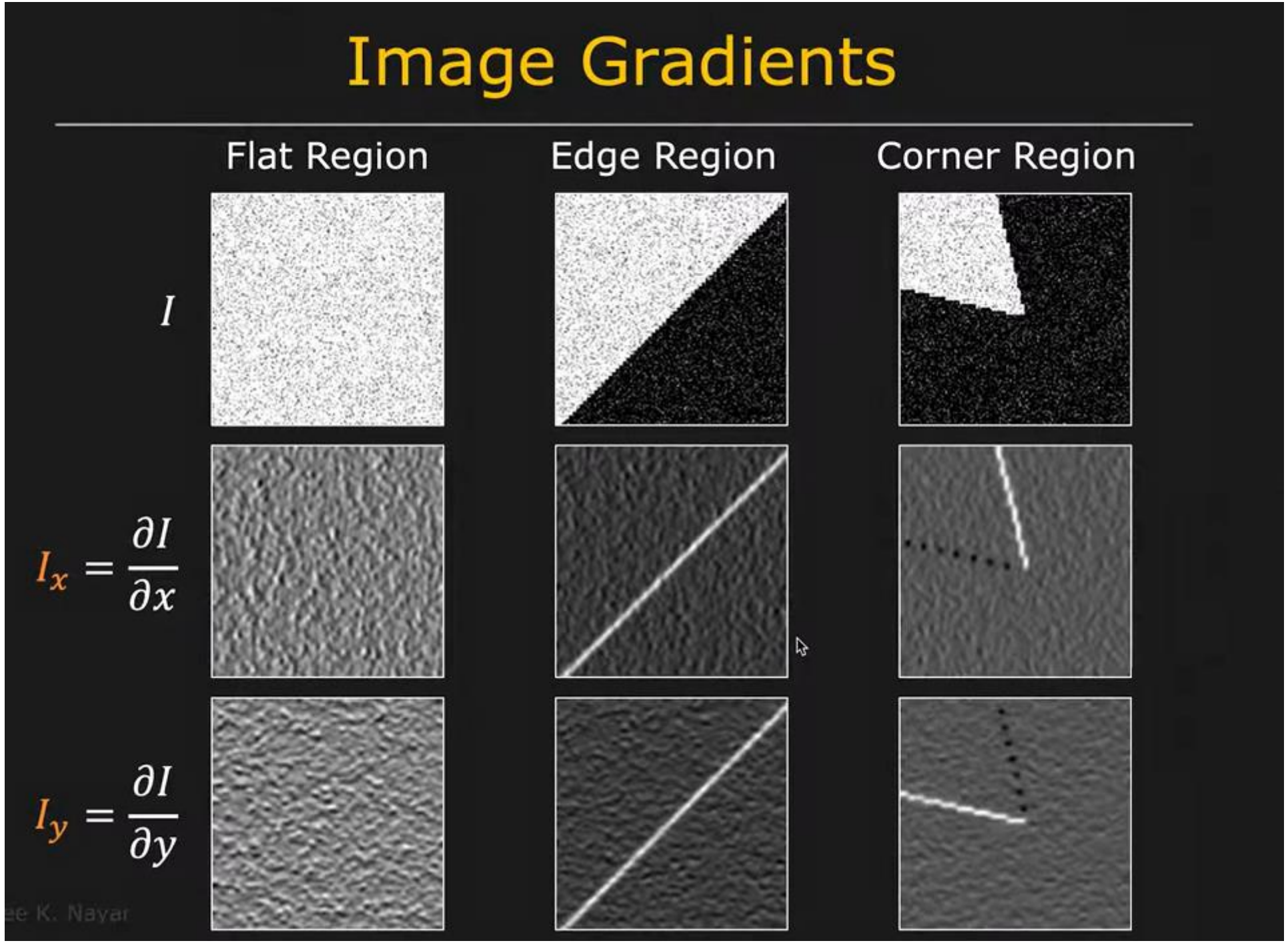


"Edge" Region



"Corner" Region

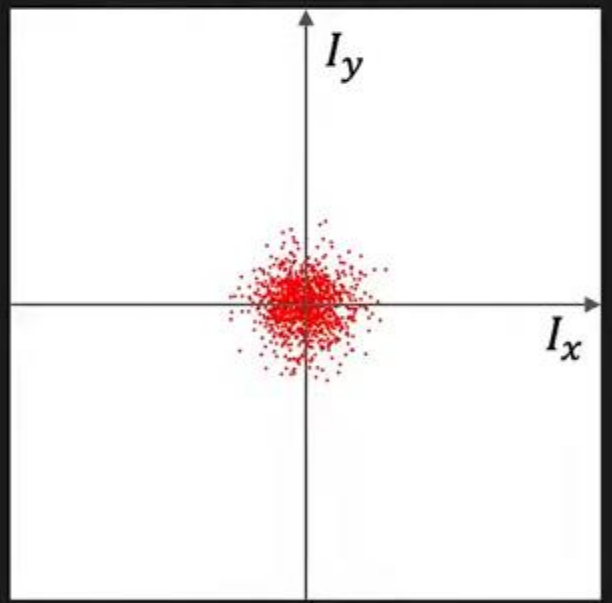
Image Gradients



Distribution of Image Gradients

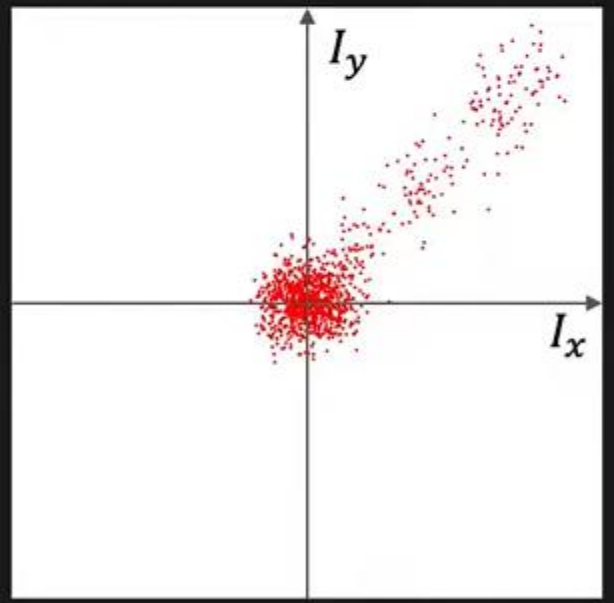
Flat Region



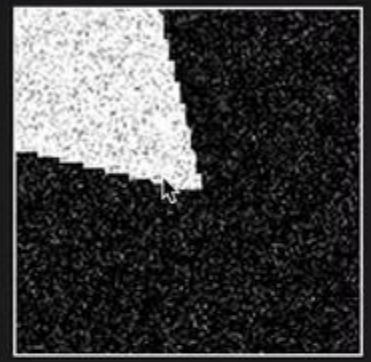


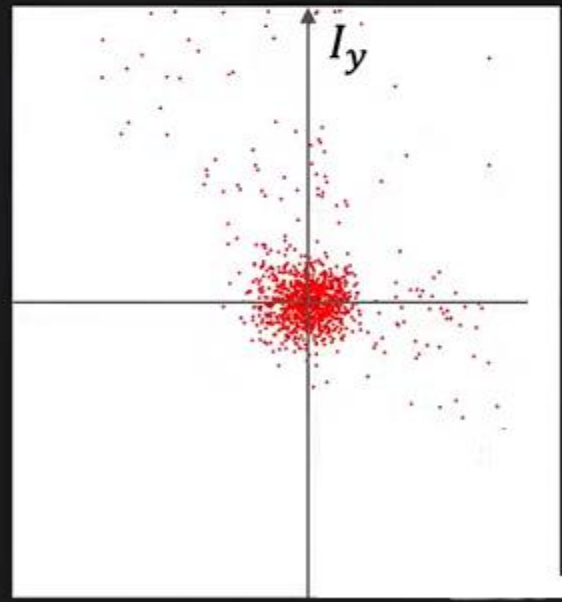
Edge Region





Corner Region

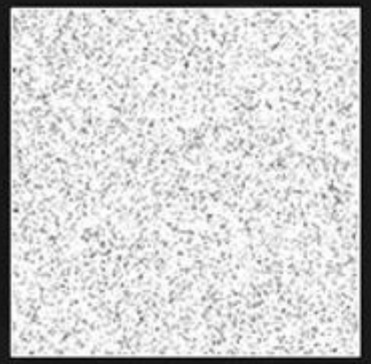




Distribution of I_x and I_y is different for all three regions.

Fitting Elliptical Disk to Distribution

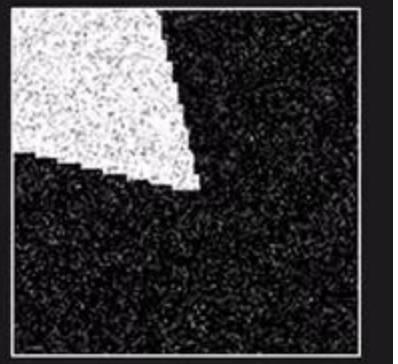
Flat Region

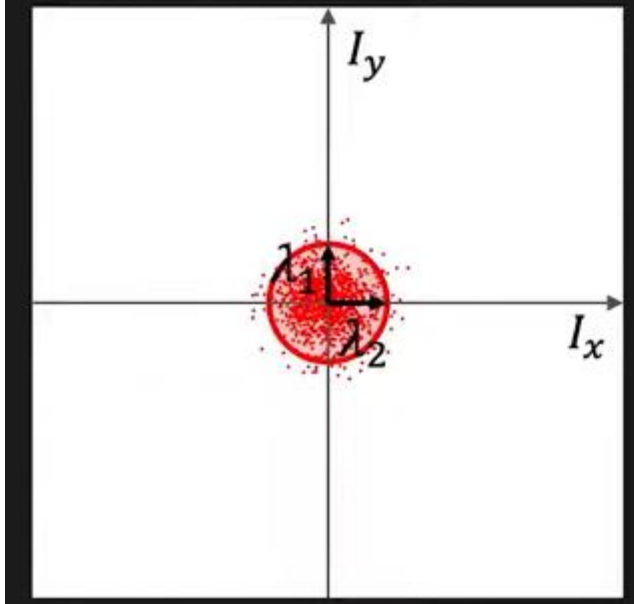


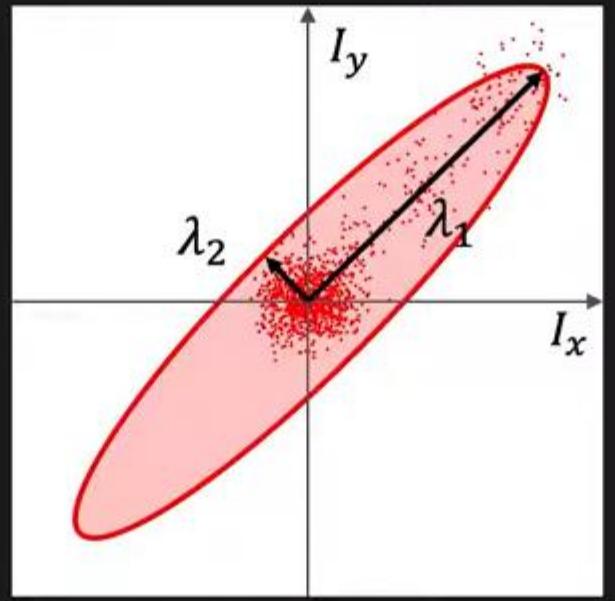
Edge Region

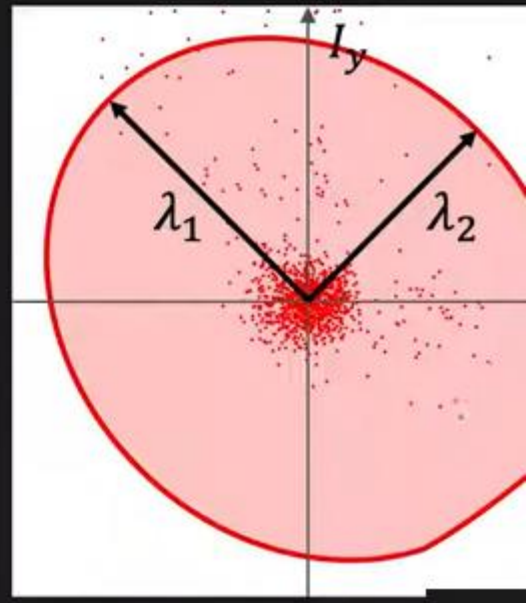


Corner Region



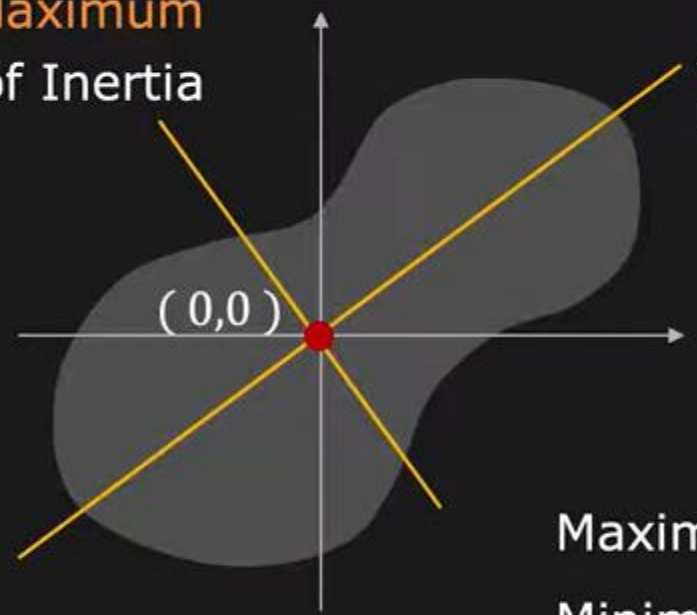






λ_1: Length of Semi-Major Axis λ_2: Length of Semi-Minor

Fitting an Elliptical Disk



Maximum Moment of Inertia = E_{max}

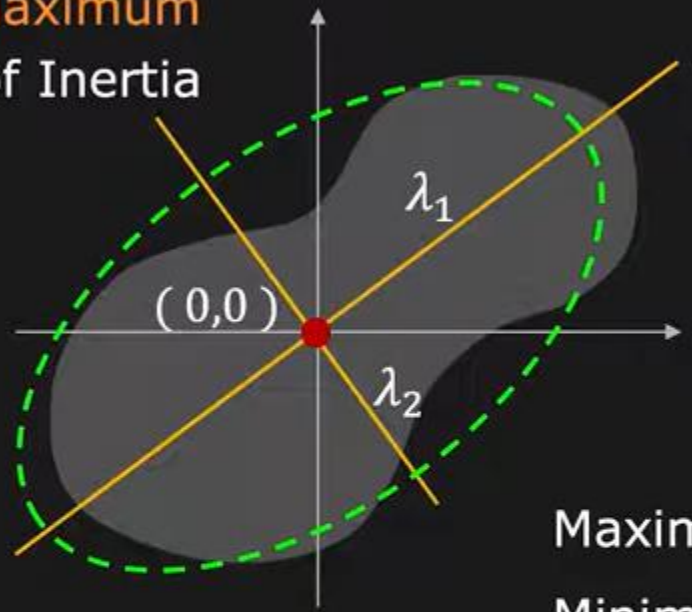
Minimum Moment of Inertia = E_{min}

(See lecture on Binary Images)

Fitting an Elliptical Disk

Axis of Maximum Moment of Inertia

Axis of Minimum Moment of Inertia



Maximum Moment of Inertia = E_{max}

Minimum Moment of Inertia = E_{min}

(See lecture on Binary Images)

Length of Semi-Major Axis = $\lambda_1 = E_{max}$

Length of Semi-Minor Axis = $\lambda_2 = E_{min}$

2020 Shree K. Nayar

Fitting an Elliptical Disk

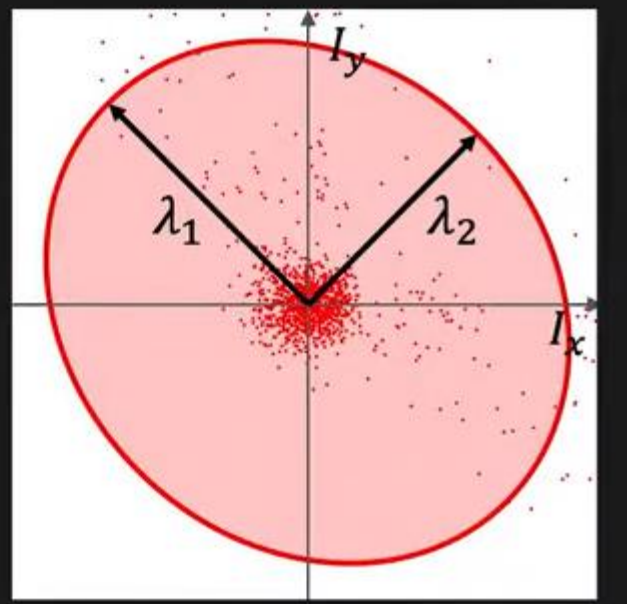
Second Moments for a Region:

$$a = \sum_{i \in W} \left({I_x}_i^2 \right) \qquad b = 2 \sum_{i \in W} \left({I_x}_i {I_y}_i \right)$$

$$c = \sum_{i \in W} \left({I_y}_i^2 \right)$$

W: Window centered at pixel

(See lecture on Binary Images)



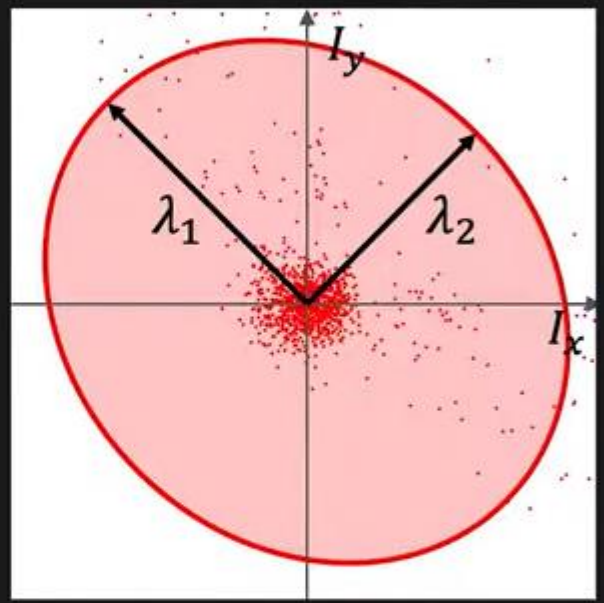
Fitting an Elliptical Disk

Second Moments for a Region:

$$a = \sum_{i \in W} \left({I_x}_i^2 \right) \qquad b = 2 \sum_{i \in W} \left(I_{x_i} I_{y_i} \right)$$

$$c = \sum_{i \in W} \left({I_y}_i^2 \right)$$ W: Window centered at pixel

(See lecture on Binary Images)



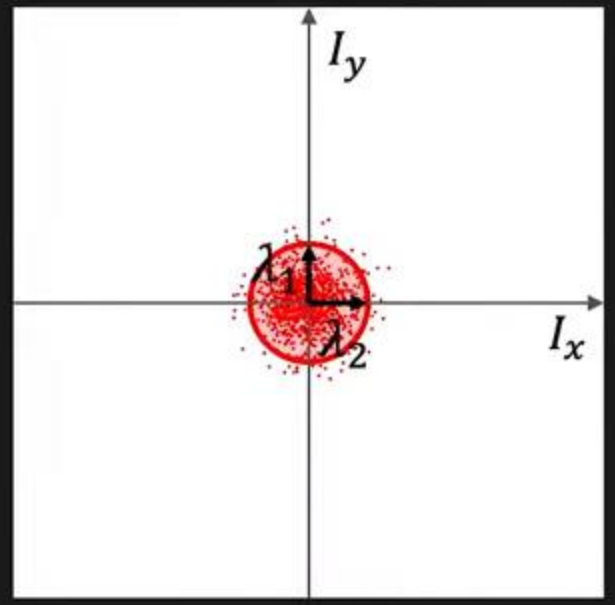
Ellipse Axes Lengths:

$$\lambda_1 = E_{max} = \frac{1}{2}\left[a + c + \sqrt{b^2 + (a-c)^2}\right]$$

$$\lambda_2 = E_{min} = \frac{1}{2}\left[a + c - \sqrt{b^2 + (a-c)^2}\right]$$

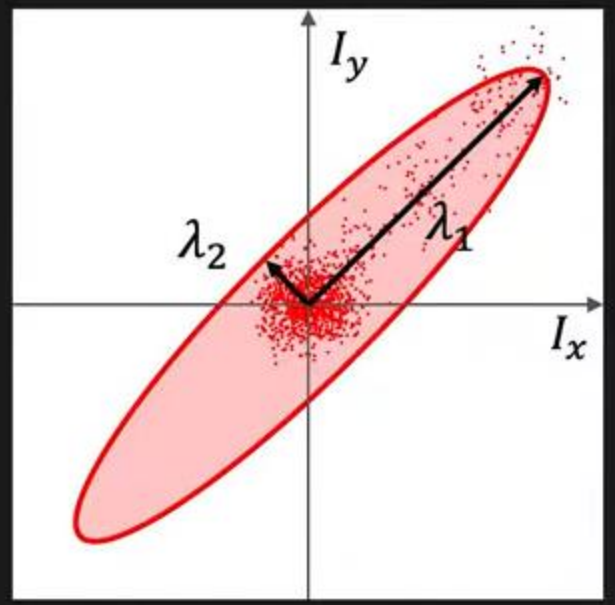
Interpretation of λ_1 and λ_2

Flat Region



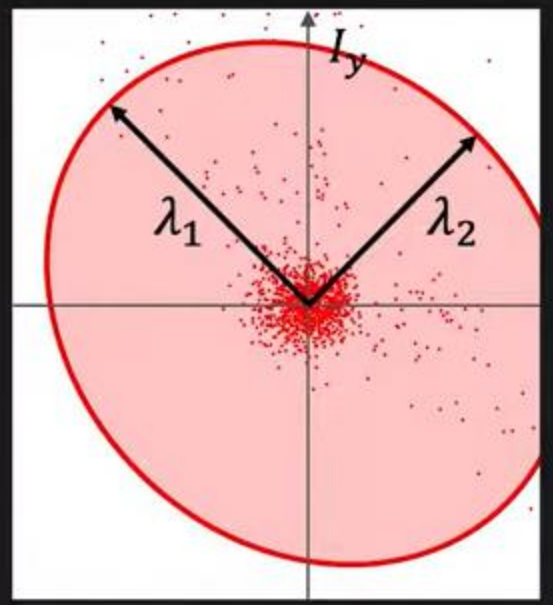
$\lambda_1 \sim \lambda_2$
Both are Small

Edge Region



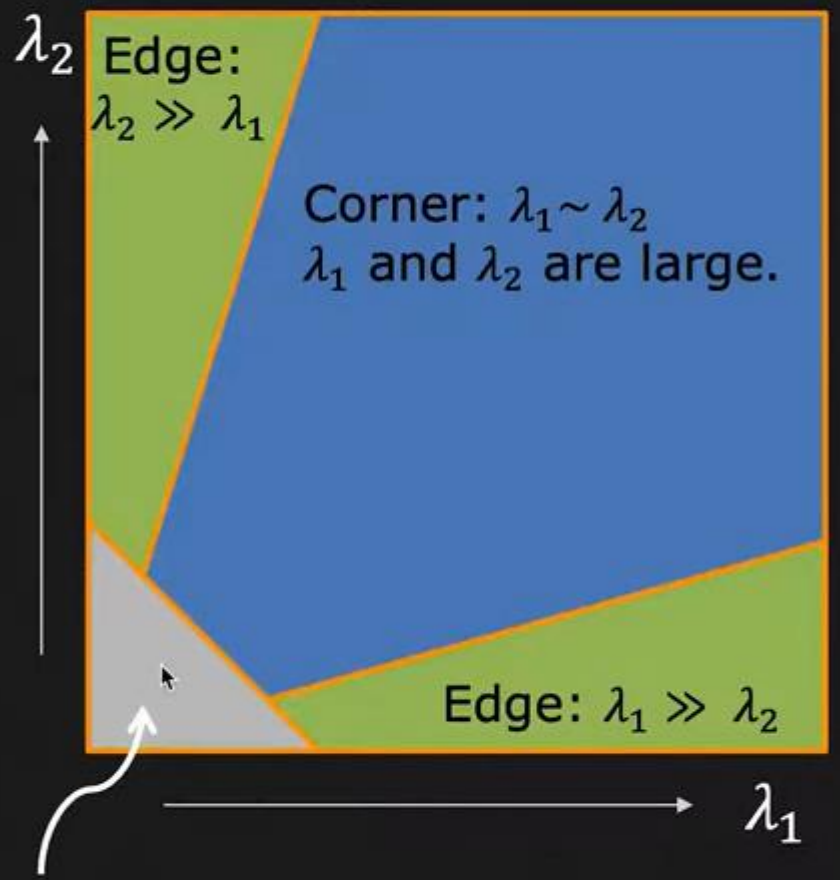
$\lambda_1 \gg \lambda_2$
λ_1 is Large
λ_2 is Small

Corner Region



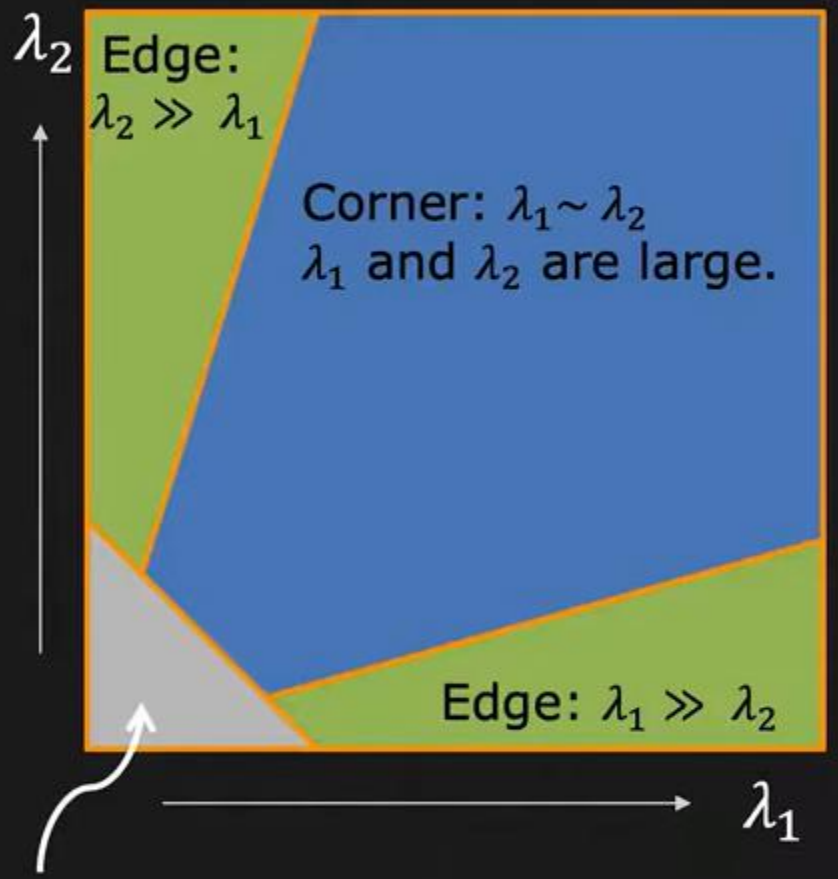
$\lambda_1 \sim \lambda_2$
Both are Large

Harris Corner Response Function

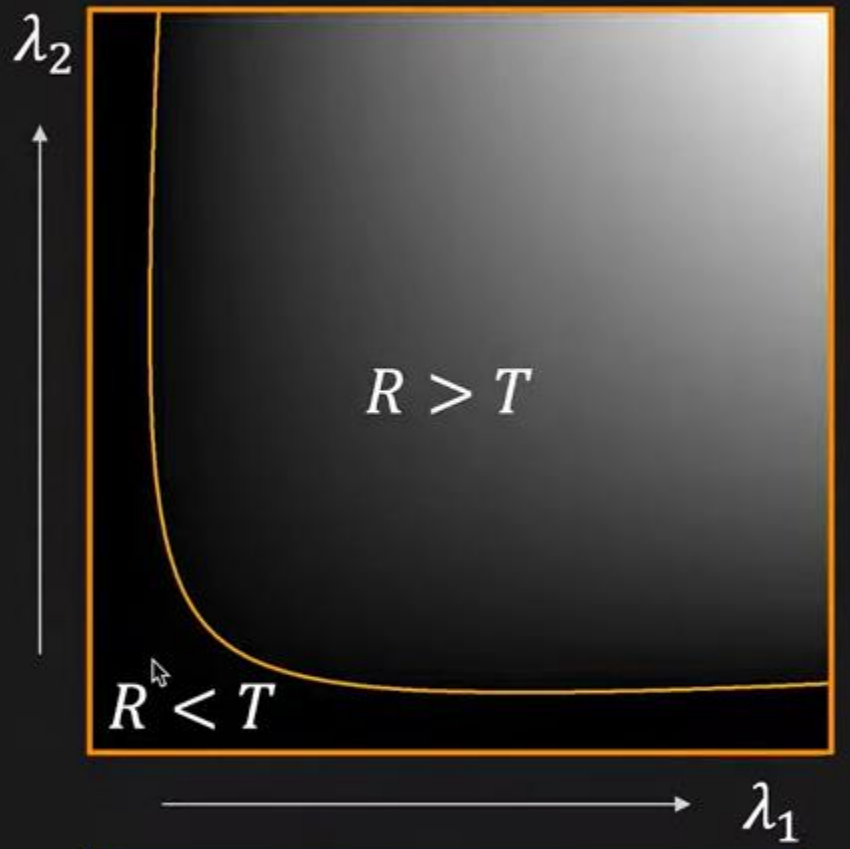


Flat: $\lambda_1 \sim \lambda_2$
λ_1 and λ_2 are small.

Harris Corner Response Function



Flat: $\lambda_1 \sim \lambda_2$
λ_1 and λ_2 are small.



$$R = \lambda_1\lambda_2 - k(\lambda_1 + \lambda_2)^2$$

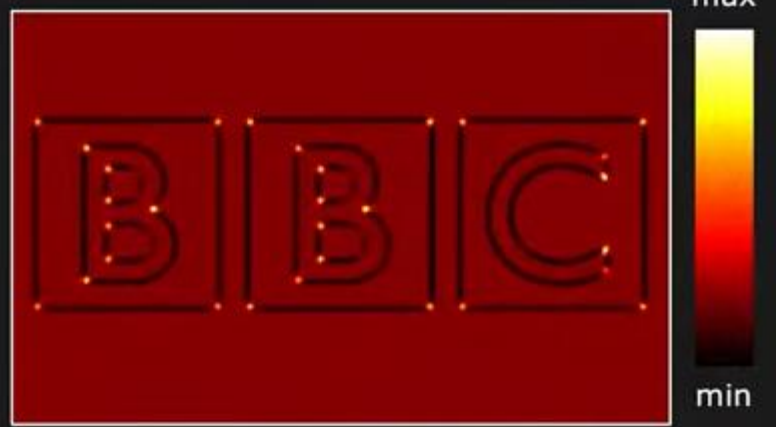
where: $0.04 \leq k \leq 0.06$

(Designed Empirically

Harris Corner Detection Example



Image



Corner Response R

Harris Corner Detection Example

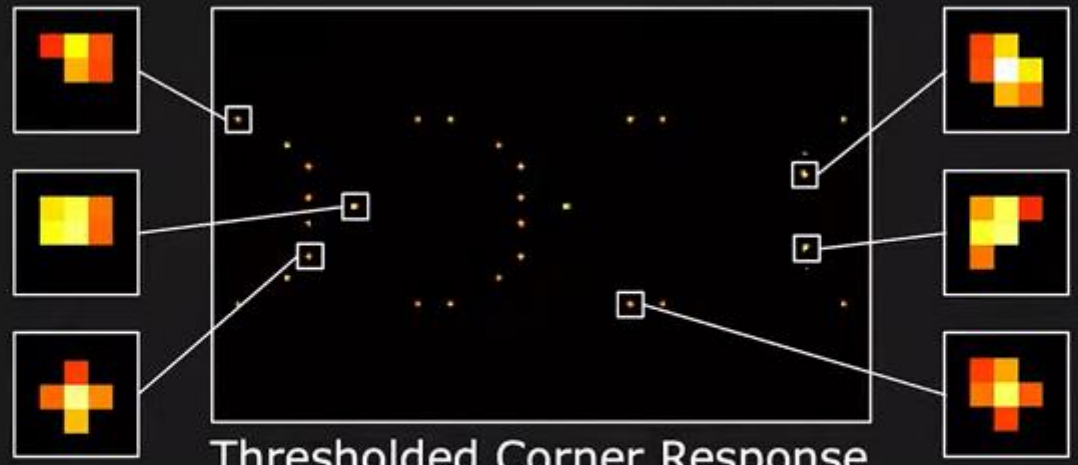


Image



Corner Response R





Thresholded Corner Response
$R > T$

How to determine the actual corner pixel?

K. Nasar

Non-Maximal Suppression

1. Slide a window of size k over the image.

2. At each position, if the pixel at the center is the maximum value within the window, label it as positive (retain it). Else label it as negative (suppress it).



Suppress Suppress Retain

Used for finding Local Extrema (Maxima/Minima)

ee K. Nayar

Harris Corner Detection Example



Image





Corner Response R



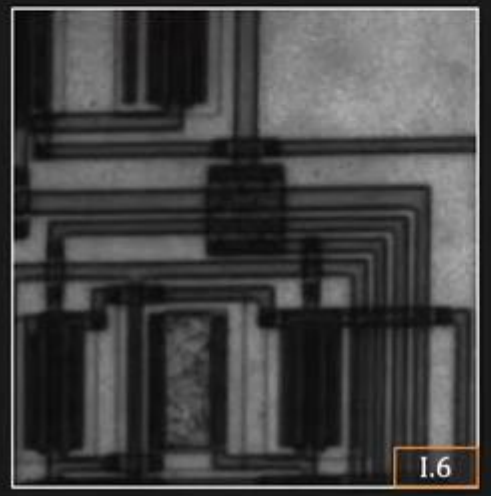
Thresholded Corner Response
$R > T$ ($T = 5.1 \times 10^7$)



Detected Corners

K. Navar

Harris Corner Detection Example



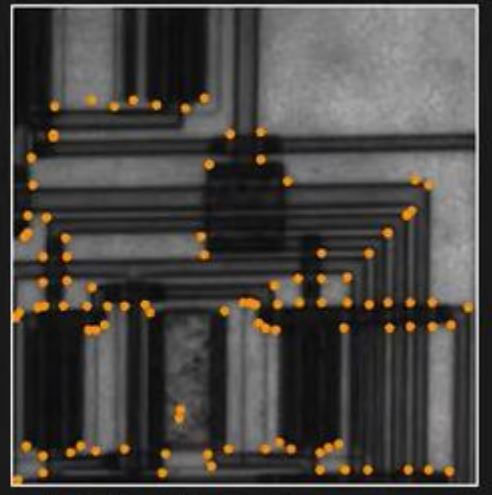
Image



Corner Response R



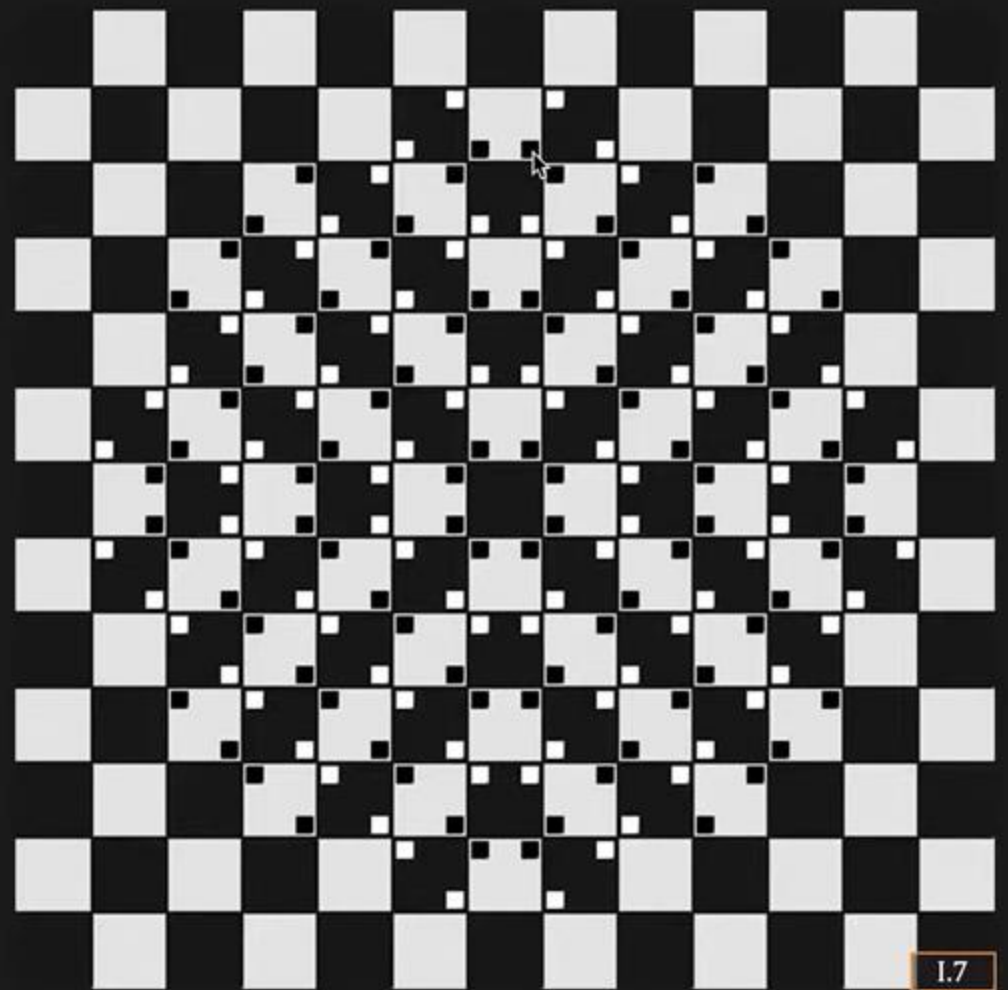
Thresholded Corner Response
$R > T$ ($T = 5.1 \times 10^7$)



Detected Corners

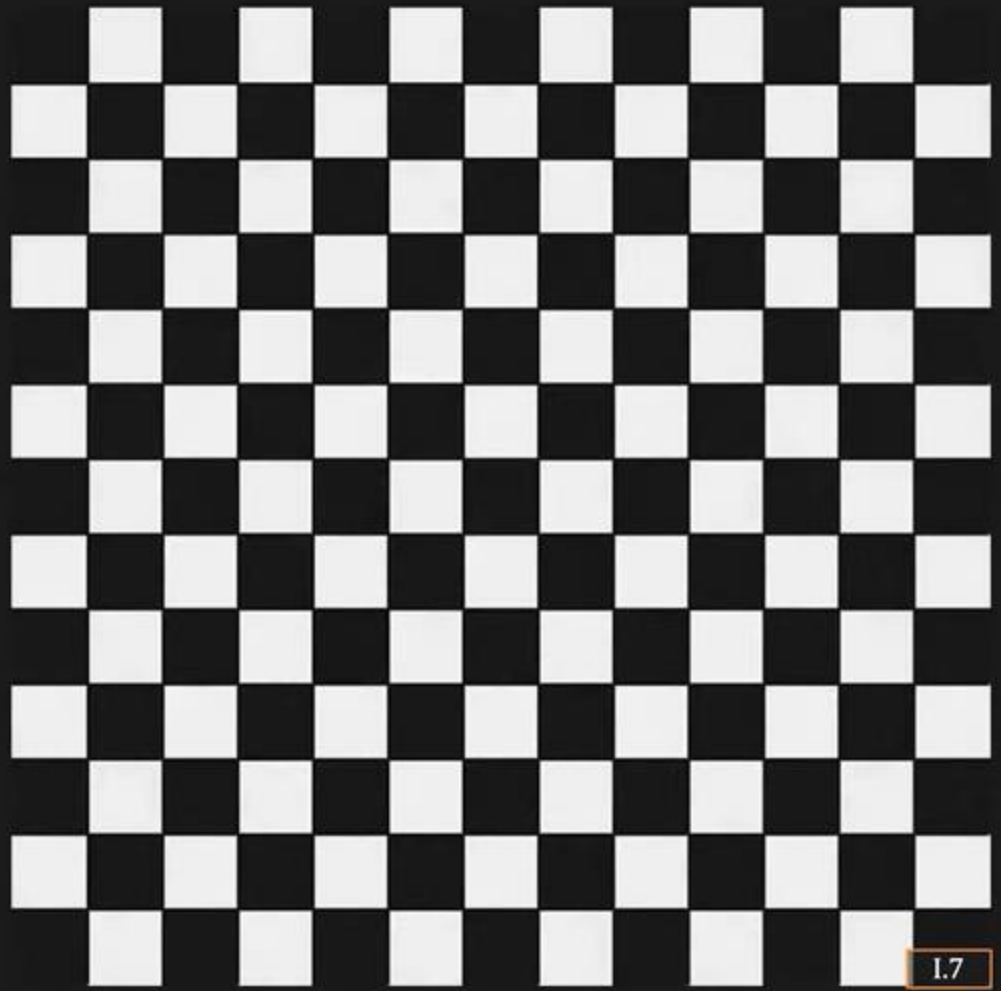
ee K. Nayar

Corner Illusion: The Bulge



Kitaoka, 1998

Corner Illusion: The Bulge



Kitaoka, 1998

Reference

- https://www.youtube.com/watch?v=Z_HwkG90Yvw